Review.

Chapter 1.

§1.1. n-dimensional vector $V = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix}$ or $V = (V_1, V_2, \cdots, V_n)$.

vector addition.

scaler multiplication.

Linear combination of vectors.

§1.2.

<u>dot product</u> or <u>inner product</u>.

$$V \cdot W = V_1 W_2 + V_2 W_2 + \cdots + V_n W_n. \rightarrow \text{scalar}$$

<u>Length</u> $\|V\| = \sqrt{V \cdot V}$

<u>Normalization</u> $u = \frac{V}{\|V\|} \longrightarrow$ unit vector

Angles between two vectors.

<u>Cosine formula</u>. $\cos\theta = \frac{V \cdot W}{\|V\| \, \|W\|}$. $\Rightarrow V \cdot W = 0$ if $V \perp W$.

<u>Schwartz inequality</u> $|V \cdot W| \leqslant \|V\| \, \|W\|$.

Chapter 2.

§2.1. $A = [a_1 \,\vdots\, a_2 \,\vdots\, \cdots \,\vdots\, a_n]$, $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$\Rightarrow Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n \longrightarrow$ multiplication by columns

$A = \begin{bmatrix} a^1 \\ a^2 \\ \vdots \\ a^n \end{bmatrix} \Rightarrow Ax = \begin{bmatrix} a^1 \cdot x \\ a^2 \cdot x \\ \vdots \\ a^n \cdot x \end{bmatrix} \longrightarrow$ multiplication by rows

Identity matrix

$$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}_{n\times n}$$

General matrix. m rows, n columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \rightarrow A = (a_{ij})_{m\times n}.$$

Transformation of a system of linear equations to $Ax = b$ (matrix form).

§2.2.

Pivot = First nonzero in the row that does the elimination.

$$\text{Multiplier} = \frac{(\text{entry to be eliminale})}{(\text{Pivot})}$$

Failure of Gaussian Elimination → Three Cases.

Gaussian elimination process.

§2.3.

Elimination matrix E_{ij} has an extra nonzero entry $-l_{ij}$ in the (i,j)-position of the Identity matrix.

⇒ $E_{ij}A$ subtracts a multiple l_{ij} of row j from row i of A.

Associative Law $A(BC) = (AB)C$

Not commutative Law $AB \neq BA$ in general

Permutation matrix P_{ij} is obtained by exchanging row i and row j of the Identity matrix.

$\Rightarrow P_{ij}A$ exchangs row i and row j of A.

<u>Gaussian elimination</u> is equivalent to apply a sequence of Elimination matrices and permution matrices to $[A\ b]$ (Augmented matrix) from the left of the linear system $AX=b$.

- Find corresponding E_{ij}'s, P_{ij}'s.

§2.4.

Matrix addition

Scalar multiplication of a matrix.

<u>Matrix multiplication</u>

$$A : m\times n \qquad B : n\times p$$

$$\rightarrow C = AB : m\times p$$

Rule: The entry in row i and column j of AB is

(row i of A) $\cdot$ (row j of B)

i.e., $C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$, $1\le i\le m$, $1\le j\le p$.

Laws for matrix operations

$$A+B = B+A$$

$$\alpha(A+B) = \alpha A + \alpha B$$

$$A+(B+C) = (A+B)+C$$

$$C(A+B) = CA+CB$$

$$(A+B)C = AC+BC$$

$$A(BC) = (AB)C$$

$$AB \neq BA \text{ in general.}$$

Power of matrix $\quad A^p = A\cdot A \cdots A$ (p factors)

Block multiplication.

$$A=\left[a_1 \,\vdots\, a_2 \,\vdots\, \cdots \,\vdots\, a_n\right], \quad B=\begin{bmatrix} b^1 \\ \hline b^2 \\ \hline \vdots \\ \hline b^n \end{bmatrix}$$

$$\Rightarrow AB=\left[a_1 b^1 + a_2 b^2 + \cdots + a_n b^n\right]$$

§2.5.

A is invertible $\Leftrightarrow$ There exists a A^{-1} such that $AA^{-1}=I=A^{-1}A$.

Not all matrices have inverse.

$(A^{-1})^{-1}=A$ if A is invertible

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{if } ad-bc \neq 0$$

$$\begin{bmatrix} d_1 & & 0 \\ & d_2 & \\ 0 & & d_n \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{d_1} & & 0 \\ & \frac{1}{d_2} & \\ 0 & & \frac{1}{d_n} \end{bmatrix} \quad \text{if } d_i \neq 0 \text{ for } i=1,\cdots,n$$

$(AB)^{-1}=B^{-1}A^{-1}$ if A and B are invertible.

$(A_1 A_2 \cdots A_m)^{-1}=A_m^{-1}\cdots A_2^{-1}A_1^{-1}$

Inverse of an Elimination matrix.

for example, $E_{ij}=\begin{bmatrix} 1 & 0 & 0 \\ -l_{ij} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow E_{ij}^{-1}=\begin{bmatrix} 1 & 0 & 0 \\ l_{ij} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

<u>Gauss-Jordan Elimination</u>. for finding A^{-1}

$$[A \ \ I] \longrightarrow [I \ \ A^{-1}]$$

§2.6.

LU factorization $A=L\,U$ → upper triangular.
L → lower triangular

3x3 matrix

$L = \begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix}$ $U \to$ remaining upper triangular matrix after Gaussian elimination. $\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix}$

LDU factorization

$$A = L\underline{DU} \longrightarrow \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix} = \begin{bmatrix} \overset{=u_{11}}{d_1} & & \\ & d_2^{=u_{22}} & \\ & & \underset{u_{33}}{\underset{\shortparallel}{d_3}} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/d_1 & u_{13}/d_1 \\ & 1 & u_{23}/d_2 \\ & & 1 \end{bmatrix}$$

$Ax = b$
$\Updownarrow$
$(LU)x = b$ $\Rightarrow$ $\begin{cases} Ly = b \to \text{using foward substitution to solve } c \\ Ux = c \to \text{then using backword substitution to solve } x. \end{cases}$

§2.7.

$A^T \to$ Transpose of A

$(A^T)_{ij} = A_{ji}$ $\quad$ $A: m \times n \to A^T: n \times m$.

Rules: $(A+B)^T = A^T + B^T$
$(AB)^T = B^T A^T$
$(A^{-1})^T = (A^T)^{-1}$

A is called a <u>symmetric matrix</u> iff <u>$A^T = A$</u>

The inverse of a symmetric matrix is also symmetric.

If $A = A^T$, $A = LDU$, then $U = L^T \Rightarrow A = LDL^T$.

<u>Permutation matrix P</u> has rows of I in any order.

P^{-1} is also a permutation matrix ; $PA \to$ row exchange of A
$P^{-1} = P^T$; $AP \to$ column exchange of A.

LU factorization with row exchange.

Chapter 3.

§ 3.1.

Definition of vector space and subspace.

two essential operations
  i) Vector addition
  ii) Scalar multiplication.

two conditions:
Let E be a set of vectors. E is a vector space/subspaces iff
  i) $V, W \in E \Rightarrow V+W \in E$
  ii) $V \in E$, α a scalar $\Rightarrow \alpha V \in E$.

Remark: All linear combinations stay in the vector space.
  Every vector space/subspace contains the zero vector.

Linear system $AX=b$.

Column space of A, denoted by $C(A)$, consists of all linear combinations of the columns of A.

$AX=b$ is solvable iff $b \in C(A)$.

If A is a $m \times n$ matrix, then $C(A)$ is a subspace of $\mathbb{R}^m$.

§ 3.2.

Defn: The null space of A denoted by $N(A)$, consists of all solutions to $AX=0$. $N(A)$ is a subspace of $\mathbb{R}^n$.

— $N(A)$ consists of all linear combinations of special solutions of $AX=0$. (solution process)

Defn: The $m \times n$ "staircase" U obtained by Gaussian elimination is called an <u>echelon matrix</u>. The reduced row echelon matrix R has zeros above the pivots as well as below and only 1's as the pivots.

§ 3.3.

Defn: The rank of a matrix A is the number of pivots and is often denoted by $r = \text{rank}(A)$.

i) $r \leq m$ and $r \leq n$.

ii) $r = m \rightarrow$ "full row rank", i.e. No zero rows in R.

$r = m \rightarrow$ "full column rank", i.e. No free variables.

Rank one matrix — every row is a multiple of the pivot row.

when $A \rightarrow R$.

i) $C(A) \neq C(R)$.

ii) $C(A^T) = C(R^T)$.

<u>Null space matrix</u> N has n rows and $n-r$ columns.

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{array}{l} \rightarrow r \text{ pivot rows} \\ \rightarrow m-r \text{ zero rows} \end{array} \rightarrow N = \begin{bmatrix} -F \\ I \end{bmatrix} \begin{array}{l} \rightarrow r \text{ pivot variables} \\ \rightarrow n-r \text{ free variables} \end{array}$$

$\downarrow$ r pivot columns, $\downarrow$ $n-r$ free variables

§ 3.4.

$[A \; b] \rightarrow [R \; d]$

$Ax = b$

complete solution: $x = x_p + x_n$

↳ The partialar solution solves $Ax_p = b$ ↳ $Rx_p = d$

The $n-r$ special solution solve $Ax_n = 0 \rightarrow Rx_n = 0$